Machine Learning

Learning from data in AWS

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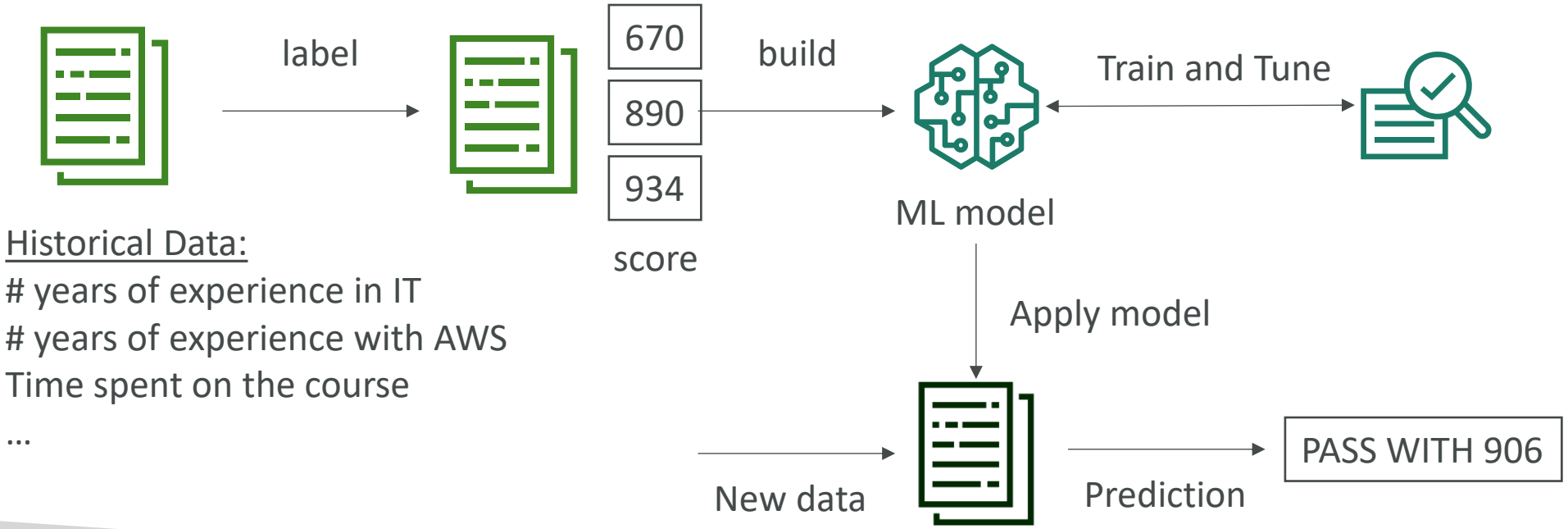
Amazon SageMaker

- Fully managed service for developers / data scientists to build ML models
- Typically, difficult to do all the processes in one place + provision servers
- Machine learning process (simplified): predicting your exam score



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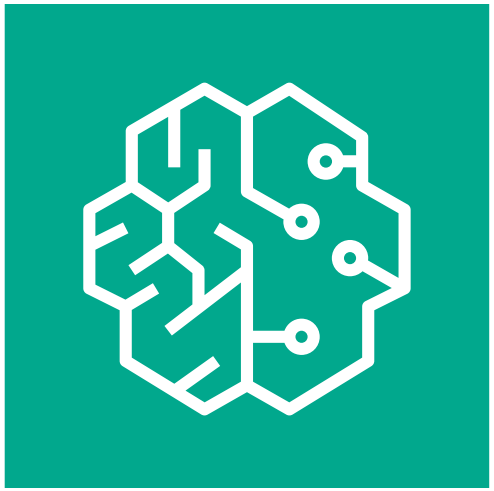
SageMaker Feature Store

- A “feature” is just a property used to train a machine learning model.
  - Like, you might predict someone’s political party based on “features” such as their address, income, age, etc.
- Machine learning models require fast, secure access to feature data for training.
- It’s also a challenge to keep it organized and share features across different models.



Amazon SageMaker Feature Store

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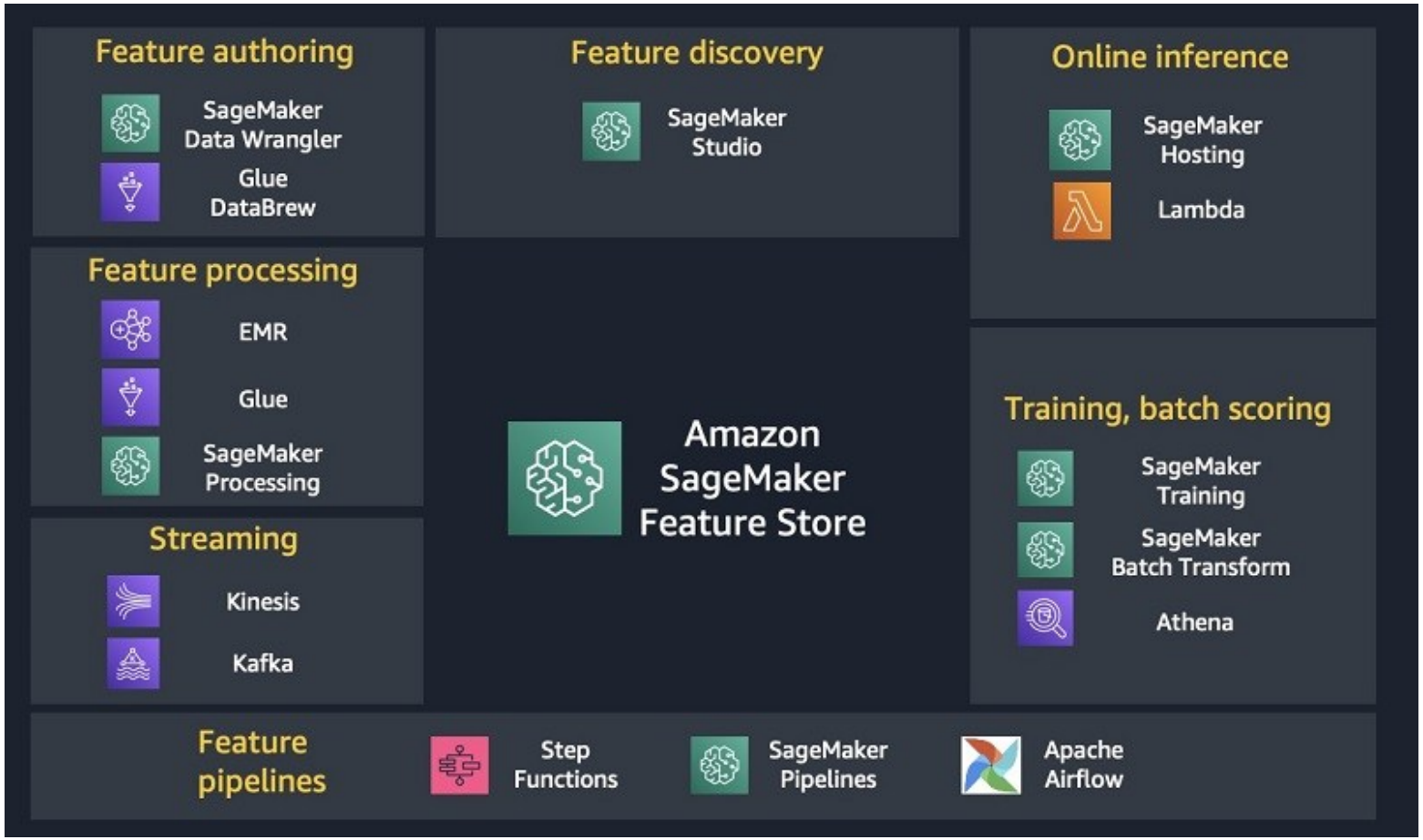
Where the features come from is up to you.



AWS

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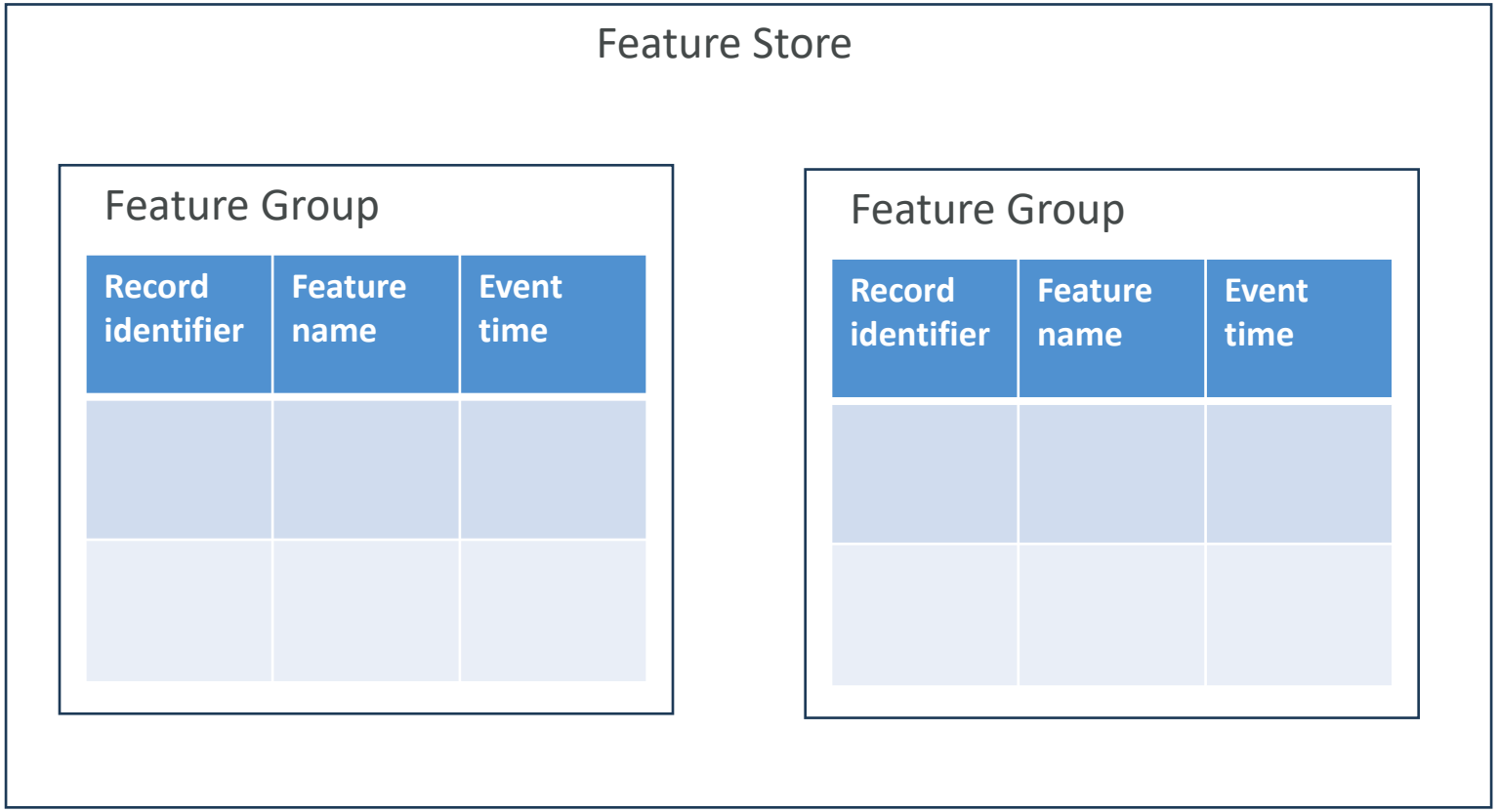
How SageMaker Feature Store Organizes Your Data



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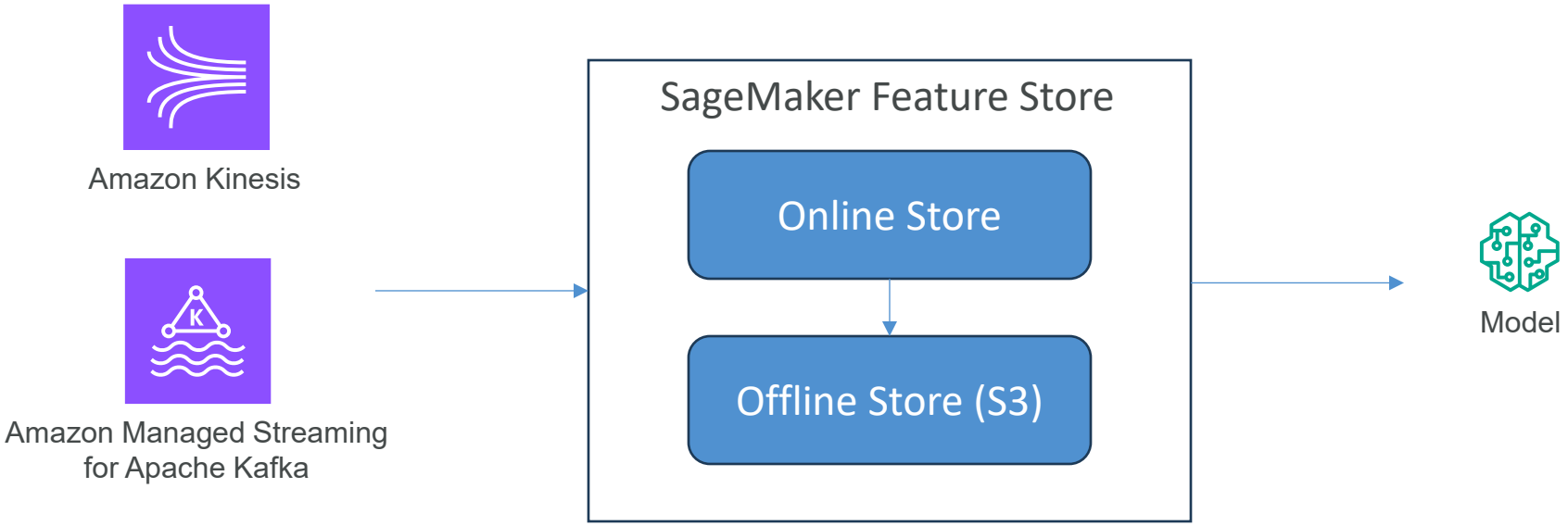
Data Ingestion (streaming or batch)



Spark, Data Wrangler, others

STREAMING access via PutRecord / GetRecord API's

BATCH access via the offline S3 store (use with anything that hits S3, like Athena, Data Wrangler. Automatically creates a Glue Data Catalog for you.)

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SageMaker Feature Store Security

- Encrypted at rest and in transit
- Works with KMS customer master keys
- Fine-grained access control with IAM
- May also be secured with AWS PrivateLink

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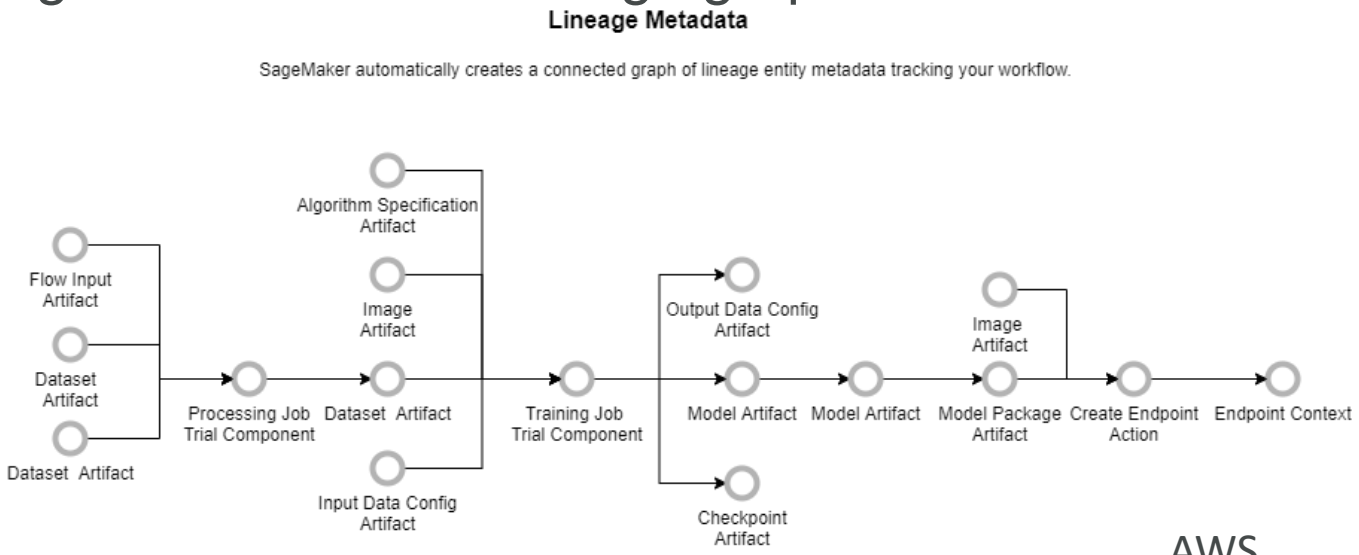
SageMaker ML Lineage Tracking

- Creates & stores your ML workflow (MLOps)
- Keep a running history of your models
- Tracking for auditing and compliance
- Automatically or manually-created tracking entities
- Integrates with AWS Resource Access Manager for cross-account lineage
- Sample SageMaker-created lineage graph:



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Lineage Tracking Entities

- Trial component (processing jobs, training jobs, transform jobs)
- Trial (a model composed of trial components)
- Experiment (a group of Trials for a given use case)
- Context (logical grouping of entities)
- Action (workflow step, model deployment
- Artifact (Object or data, such as an S3 bucket or an image in ECR)
- Association (connects entities together) – has optional AssociationType:
  - ContributedTo
  - AssociatedWith
  - DerivedFrom
  - Produced
  - SameAs

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Querying Lineage Entities

- Use the LineageQuery API from Python
  - Part of the Amazon SageMaker SDK for Python
- Do things like find all models / endpoints / etc. that use a given artifact
- Produce a visualization
  - Requires external Visualizer helper class

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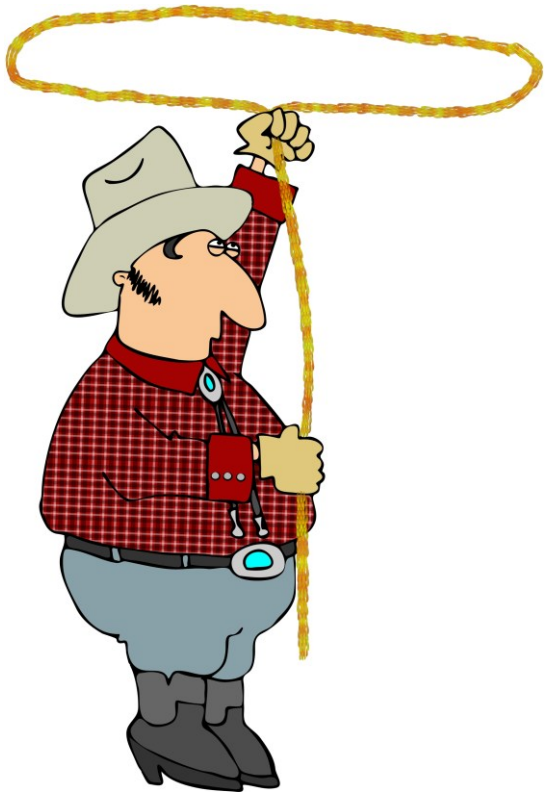
SageMaker Data Wrangler

- Visual interface (in SageMaker Studio) to prepare data for machine learning
- Import data
- Visualize data
- Transform data (300+ transformations to choose from)
  - Or integrate your own custom xforms with pandas, PySpark, PySpark SQL
- “Quick Model” to train your model with your data and measure its results



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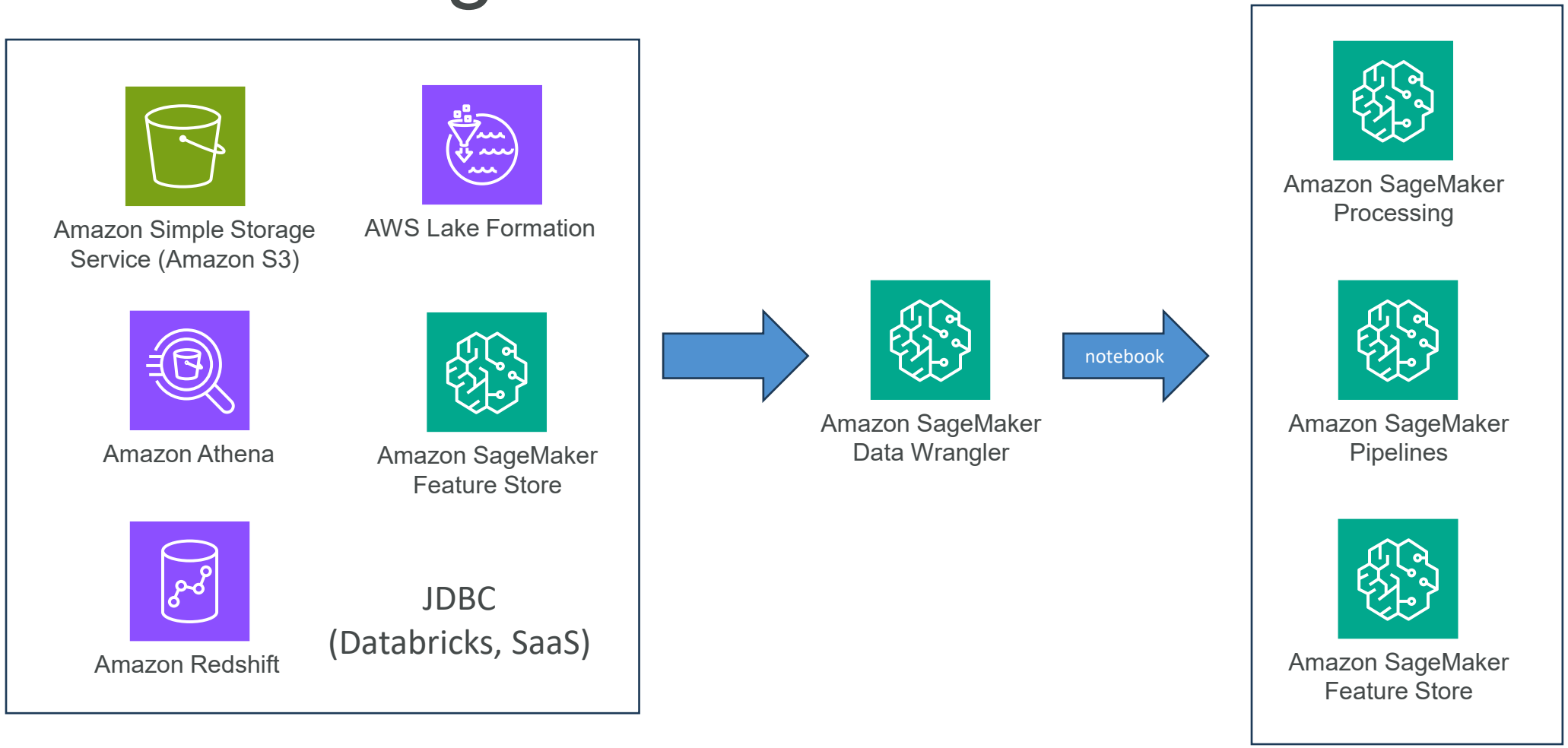
Data Wrangler sources



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Data Wrangler: Import Data

Amazon SageMaker Studio | File | Edit | View | Run | Kernel | Git | Tabs | Settings | Help | Feedback

/ Untitled Folder /

Name

titanic.flow

titanic.flow

Import | Prepare | Analyze | Export

Data sources / S3 source / sagemaker-us-east-2-613904931467 / titanic

Import a dataset from S3

Use the following table to browse S3. Select a file to see import options. The following file formats are supported: CSV and Parquet.

Object Name	Size	Last Modified
titanic-train.csv	58.89KB	2020-10-15 08:49:45+00:00

Previous | Displaying 1 - 1 | Next

Preview

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp
1	0	3	Braund, Mr. Owen Harris	male	22	1
2	1	1	Cumings, Mrs. John Bra...	female	38	1
3	1	3	Heikkinen, Miss. Laina	female	26	0
4	1	1	Futrelle, Mrs. Jacques H...	female	35	1
5	0	3	Allen, Mr. William Henry	male	35	0

DETAILS

Name: titanic-train.csv (*Required*)

URI: s3://sagemaker-us-east-2-[redacted]

File type: csv (*Required*)

☑ Add header to table

☑ Enable sampling

Import dataset

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Data Wrangler: Preview Data

Amazon SageMaker Studio | File | Edit | View | Run | Kernel | Git | Tabs | Settings | Help | Feedback

/ Untitled Folder /

Name

titanic.flow

titanic.flow

Import | Prepare | Analyze | Export

Data flow / Transform: titanic-train.csv

Configure types

PassengerId (long)	Survived (long)	Pclass (long)	Name (string)	Sex (string)	Age (long)
7	0	1	McCarthy, Mr. Timothy J	male	54
8	0	3	Palsson, Master. Gosta ...	male	2
9	1	3	Johnson, Mrs. Oscar W (...	female	27
10	1	2	Nasser, Mrs. Nicholas (A...	female	14
11	1	3	Sandstrom, Miss. Margu...	female	4
12	1	1	Bonnell, Miss. Elizabeth	female	58
13	0	3	Saundercock, Mr. Willia...	male	20
14	0	3	Andersson, Mr. Anders J...	male	39
15	0	3	Vestrom, Miss. Hulda A...	female	14
16	1	2	Hewlett, Mrs. (Mary D K...	female	55
17	0	3	Rice, Master. Eugene	male	2
18	1	2	Williams, Mr. Charles Eu...	male	
19	0	3	Vander Planke, Mrs. Juli...	female	31
20	1	3	Masselmani, Mrs. Fatima	female	
21	0	2	Fynney, Mr. Joseph J	male	35
22	1	2	Beesley, Mr. Lawrence	male	34
23	1	3	McGowan, Miss. Anna "...	female	15
24	1	1	Sloper, Mr. William Tho...	male	28
25	0	3	Palsson, Miss. Torborg ...	female	8
26	1	3	Asplund, Mrs. Carl Osca...	female	38
27	0	3	Emir, Mr. Farred Chehab	male	
28	0	1	Fortune, Mr. Charles Ale...	male	19

Back to data flow

CONFIGURE TYPES

Column name	Type
PassengerId	Long
Survived	Long
Pclass	Long
Name	String
Sex	String
Age	Long
SibSp	Long
Parch	Long
Ticket	String
Fare	Float
Cabin	String
Embarked	String

Clear | Preview | Apply

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Data Wrangler: Visualize Data

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/ Untitled Folder /

Name

titanic.flow

titanic.flow

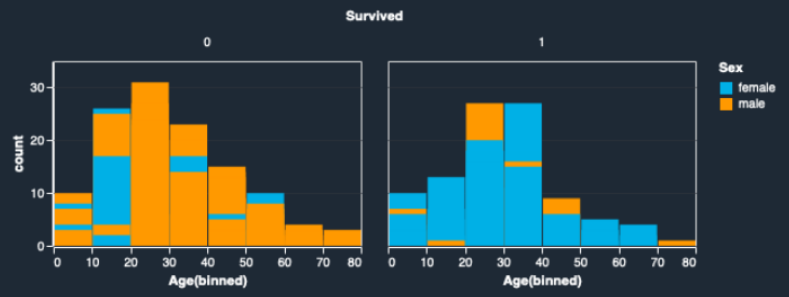
Import | Prepare | Analyze | Export

Imported datasets / Transform: titanic-train.csv / Survived vs Age

Create Analysis

Create an analysis of your data. Learn more.

Histogram: Survived vs Age



Data table

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp
1	0	3	Braund, Mr. Owen Harris	male	22	1
2	1	1	Cumings, Mrs. John Bra...	female	38	1
3	1	3	Heikkinen, Miss. Laina	female	26	0
4	1	1	Futrelle, Mrs. Jacques H...	female	35	1

Back to all analyses

Configure | Code

Chart: Histogram

Name: Survived vs Age
Optional

X axis: Age

Color by: Sex
Optional

Facet by: Survived
Optional

Cancel | Create

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Data Wrangler: Transform Data

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/ Untitled Folder /

Name

titanic.flow

titanic.flow

Import | Prepare | Analyze | Export

Data flow / Transform: titanic-train.csv

Previewing Encode categorical

Transform: titanic-train.csv

t)	Cabin (string)	Embarked (string)	Pclass__3 (float)	Pclass__1 (float)	Pclass__2 (float)
		S	1	0	0
	C85	C	0	1	0
		S	1	0	0
	C123	S	0	1	0
		S	1	0	0
		Q	1	0	0
	E46	S	0	1	0
		S	1	0	0
		S	1	0	0
		C	0	0	1
	G6	S	1	0	0
	C103	S	0	1	0
		S	1	0	0
		S	1	0	0
		S	1	0	0
		S	0	0	1
		Q	1	0	0
		S	0	0	1
		S	1	0	0

Back to data flow

Custom formula

Encode categorical

Encode a categorical variable Learn more.

Input column: Pclass

Output column name: Pclass_

Transform: One-hot encode

Invalid handling strategy: Keep

Optional

Drop last category: *Select...*

Optional

Is input ordinal encoded? *Select...*

Optional

Output format: Flattened

Optional

Clear | Preview | Add

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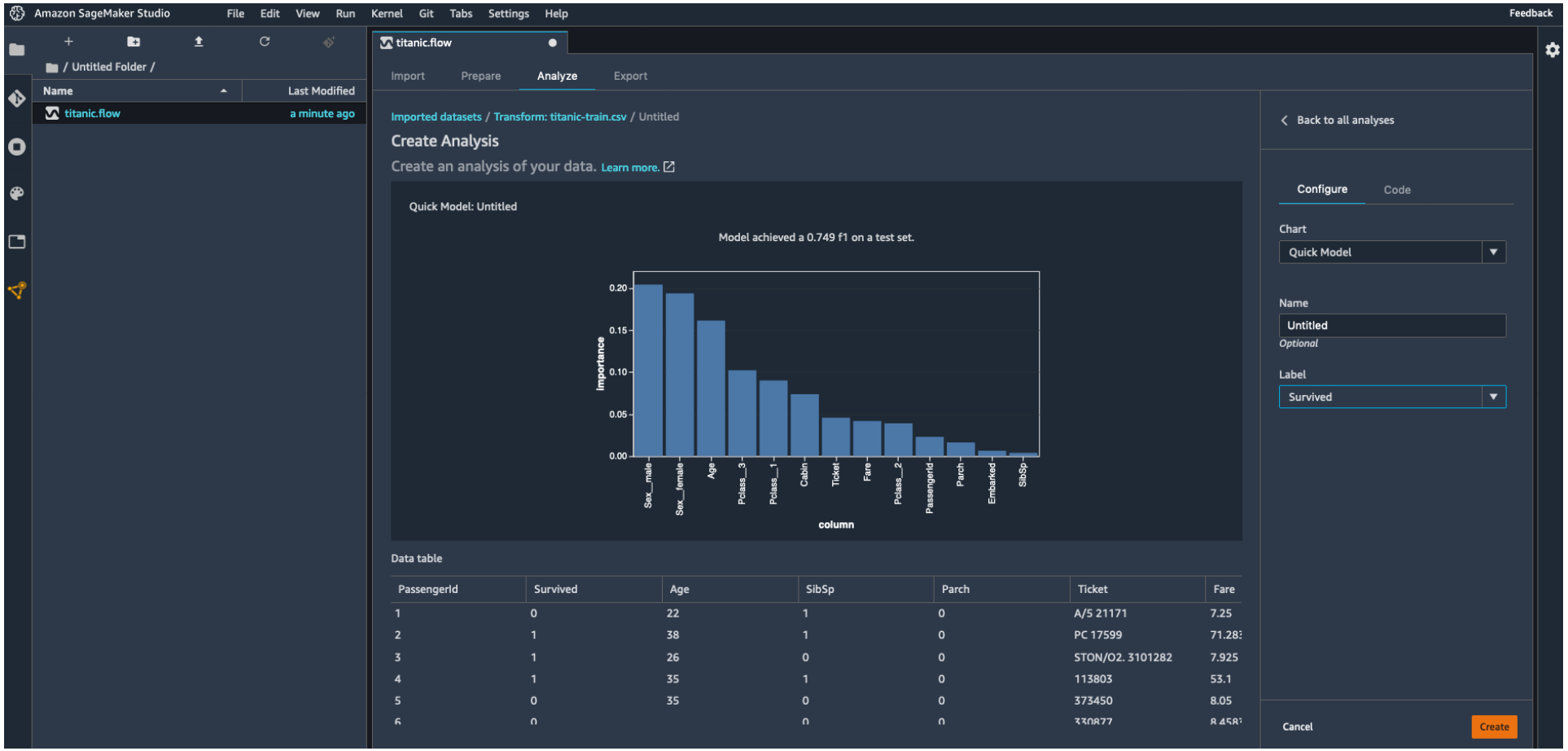
Data Wrangler: Quick Model



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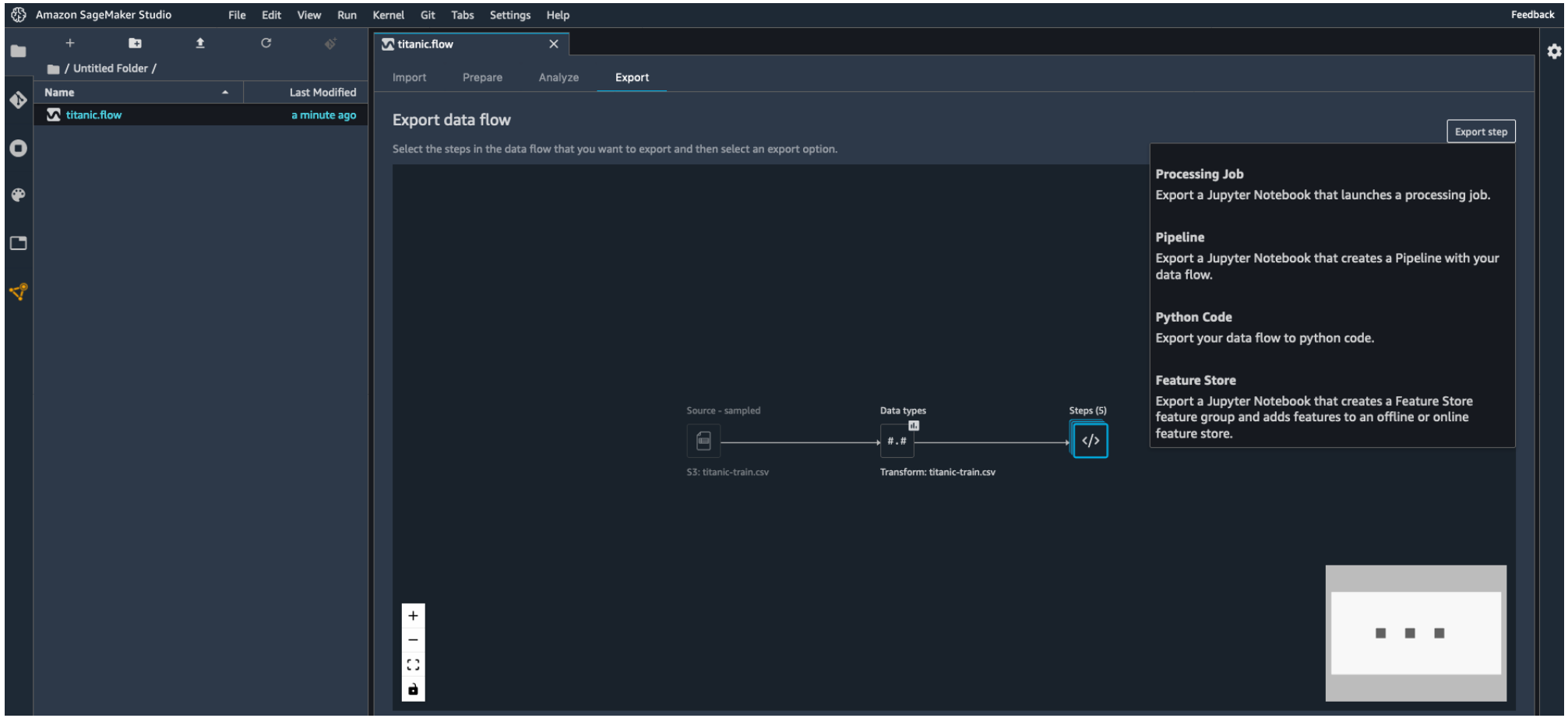
Data Wrangler: Export Data Flow



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Data Wrangler Troubleshooting

- Make sure your Studio user has appropriate IAM roles
- Make sure permissions on your data sources allow Data Wrangler access
  - Add AmazonSageMakerFullAccess policy
- EC2 instance limit
  - If you get “The following instance type is not available…” errors
  - May need to request a quota increase
  - Service Quotas / Amazon SageMaker / Studio KernelGateway Apps running on ml.m5.4xlarge instance

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